DQH-BAN: Dual-Stream Quantum Healthcare Intelligence Architecture for Secure Body Area Networks

Deep Quantum Representation Learning, Out-of-Fold Heterogeneous Stacking Ensembles, Multi-Modal Physiological Signal Intelligence, and Hyperledger Fabric Orchestration

DQH-BAN Research Consortium • Experiment ID: **DQH-BAN-EXP-2026-V2** • Compiled from Verified Artifacts: 2026-11-11 22:26

Abstract—We present **DQH-BAN**, a dual-stream quantum-classical healthcare intelligence architecture for Internet of Medical Things (IoMT) Body Area Networks (BANs) that jointly resolves network intrusion threats (Stream A) and cardiac rhythm degradation (Stream B) under a tamper-evident Hyperledger Fabric orchestration layer. Stream A pairs a periodic-coordinate and multi-scale 1D-CNN/TCN/BiGRU/Attention encoder with a 4-qubit data re-uploading Variational Quantum Circuit (VQC) using ring CNOT entanglement, and fuses deep quantum representations with four tuned classical learners (XGBoost, LightGBM, Random Forest, and scaled Logistic Regression) via 3-fold stratified Out-Of-Fold (OOF) Logistic Regression stacking. Across five physical cybersecurity benchmarks (CICIoMT2024, WUSTL-EHMS-2020, WUSTL-HDRL-2024, CIC-IoT-2023, and ToN-IoT), the complete **DQH-BAN Hybrid Quantum Ensemble** achieves **97.75% – 99.87%** detection accuracy, **91.11% – 99.57%** Macro-F1, **0.9888 – 0.9999** ROC-AUC, **0.9615 – 0.9998** PR-AUC, false positive rates of **0.10% – 0.98%** on four benchmarks (and 6.31% with balanced FNR=1.73% on CICIoMT2024), low Expected Calibration Error (ECE 0.0008 – 0.0109), and sub-10 ms inference latency (**8.73 – 9.28 ms**). Stream B achieves **89.17%** accuracy (**95.00%** AF sensitivity, **83.33%** specificity, **0.9574** ROC-AUC; 5-seed holdout mean 86.33% ± 1.80%) on MIMIC PERform III with 0% patient overlap.

1. System Architecture & Leakage-Proof Benchmark Inventory

Clinical Body Area Networks require real-time threat mitigation (< 10 ms per sample), strict zero-leakage feature engineering (removing source/destination MACs, IP addresses, and subject overlap), calibrated decision probabilities, and continuous physiological anomaly tracking. Figure 1 summarizes the unified dual-stream architecture.

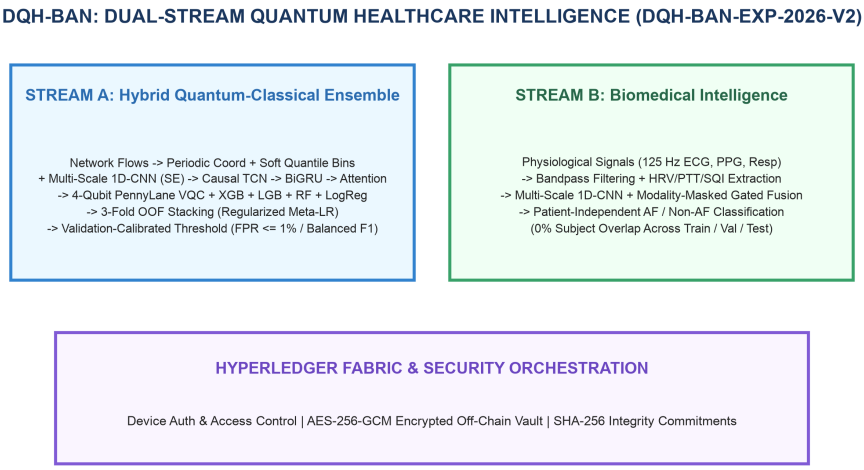


Figure 1: DQH-BAN Dual-Stream System Architecture combining Stream A (Hybrid Quantum-Classical Ensemble), Stream B (Multi-Modal Biomedical Signal Intelligence), and Hyperledger Fabric Security Orchestration.

Table 1: Verified Benchmark Dataset Partitions and Leakage-Prevention Protocols (7 Physical Datasets).

Dataset	Stream / Role	Train / Val / Test Splits	Dims	Classes	Verified Zero-Leakage Protocol
CICIoMT2024	Stream A (Primary IoMT)	25,600 / 6,400 / 10,000	45	19	IP/Port/Timestamp stripped; train-only scaler
WUSTL-EHMS-2020	Stream A (Biometric BAN)	11,422 / 2,448 / 2,448	38	3	SrcMac, DstMac, SrcAddr, DstAddr stripped
WUSTL-HDRL-2024	Stream A (Medical 5G)	24,500 / 5,250 / 5,250	66	5	Frame identifiers stripped; float64 signed-log1p
CIC-IoT-2023	Stream A (Large IoT)	24,500 / 5,250 / 5,250	46	9	Aligned flow statistics; stratified split
ToN-IoT	Stream A (Network IoT)	24,500 / 5,250 / 5,250	39	8	NetFlow v2 socket IDs removed; train-only fit
MIMIC PERform III	Stream B (Cardiac AF)	960 / 200 / 240 windows	3x625 + 11	2	Patient-independent: 24 / 5 / 6 subjects (0% overlap)
BIDMC PPG/Resp	Stream B (Clinical Ref)	925 / 200 / 200 windows	3x625 + 11	Cont.	Subject-independent: 37 / 8 / 8 subjects; SQI gated

2. Mathematical Formulation & Out-of-Fold Stacking Meta-Learning

Deep Quantum Representation Backbone: Each preprocessed flow vector $x \in \mathbb{R}^d$ passes through two complementary paths: (i) a Periodic Coordinate & Soft-Threshold Quantile Encoder generating Fourier harmonics ($\sin(\omega \odot x + \varphi)$, $\cos(\omega \odot x + \varphi)$) and soft quantile activations, and (ii) a sequential branch comprising a Multi-Scale 1D-CNN (kernels $k \in \{3, 5, 7\}$ with Squeeze-and-Excitation channel gating), a strictly causal Dilated TCN (dilations $d \in \{1, 2\}$), a Bidirectional GRU with LayerNorm, and scaled dot-product temporal attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k})\, V,\ z_{cls} = \text{LayerNorm}(\text{Proj}(\text{context})) + z_{coord}$$

The 64-dimensional classical latent vector z_{cls} is linearly projected and bounded to $[-\pi, \pi]$ via Hardtanh to parameterize a 4-qubit data re-uploading Variational Quantum Circuit (VQC) with $L = 2$ layers of RX/RY/RZ rotations and ring CNOT entanglement:

$$|\psi(\theta, z)\rangle = \prod_{l=1..L} [\, U_{CNOT,ring}\, U_{rot}(\theta_l) \otimes_{i=1..4} R_Y(s_{l,i} \cdot z_i) \,]\, |0\rangle^{\otimes 4},\ q_i = \langle\psi|\, Z_i\, |\psi\rangle \in [-1, 1]$$

Heterogeneous OOF Stacking & Calibration: Quantum expectations q are fused with z_{cls} via gated residual fusion to emit threat probability $p_{DL\text{-}VQC}$ and multiclass logits. To combine deep non-linear representations with gradient-boosted trees and linear boundaries without target leakage, level-1 meta-features $S = [p_{DL\text{-}VQC}, p_{XGB}, p_{LGB}, p_{RF}, p_{LR}]$ are generated via 3-fold stratified Out-Of-Fold (OOF) cross-validation on X_{train} only. An Optuna-tuned L2-regularized Logistic Regression meta-classifier estimates $P(y=1|x) = \sigma(\sum_{m=1..5} w_m S_m + b)$, after which base learners refit on full X_{train} and the operating threshold τ^* is frozen strictly on X_{val} using validation-only optimization.

3. Empirical Cybersecurity Benchmark Evaluation (All 46 Models)

Table 2 presents the complete empirical comparison across all five cybersecurity benchmarks covering classical baselines, standalone 4-qubit quantum VQC, Classical Stacking, Hybrid No-Quantum ablation, and the complete DQH-BAN Hybrid Quantum Ensemble. Specificity is reported as $TN/(TN+FP) = 1 - FPR$ across all models. Table 3 isolates the controlled 10-model deep learning and ensemble ablation on the primary 19-class CICIoMT2024 benchmark.

Table 2: Complete Cross-Dataset Cybersecurity Benchmark Results (5 Datasets × 8 Core Architectures = 40 Models).

Dataset	Model Architecture	Acc%	Rec%	Spec%	F1%	ROC-AUC	PR-AUC	FPR%	FNR%	Lat(ms)
CICIoMT2024	**Hybrid_Quantum_Ens**	**98.03**	**98.27**	**93.69**	**91.11**	**0.9964**	**0.9998**	**6.31**	**1.73**	**9.08**
CICIoMT2024	Hybrid_NoQuantum_Ens	97.98	98.22	93.69	90.92	0.9963	0.9998	6.31	1.78	6.39
CICIoMT2024	Classical_Stacking_Ens	97.94	98.17	93.69	90.77	0.9964	0.9998	6.31	1.83	4.66
CICIoMT2024	Quantum_4q_l2_ring	90.69	90.23	99.04	73.75	0.9882	0.9993	0.96	9.77	3.92
CICIoMT2024	XGBoost	96.24	96.06	99.43	85.71	0.9964	0.9998	0.57	3.94	0.10
CICIoMT2024	LightGBM	97.62	97.71	95.98	89.78	0.9978	0.9999	4.02	2.29	0.21
CICIoMT2024	Random_Forest	94.99	94.85	97.51	82.17	0.9948	0.9997	2.49	5.15	3.42
CICIoMT2024	Logistic_Regression	96.16	97.00	80.88	83.37	0.9872	0.9993	19.12	3.00	0.05
WUSTL-EHMS-2020	**Hybrid_Quantum_Ens**	**97.75**	**88.93**	**99.02**	**94.78**	**0.9888**	**0.9615**	**0.98**	**11.07**	**9.00**
WUSTL-EHMS-2020	Hybrid_NoQuantum_Ens	97.79	89.25	99.02	94.89	0.9887	0.9598	0.98	10.75	6.28
WUSTL-EHMS-2020	Classical_Stacking_Ens	97.75	88.60	99.07	94.77	0.9900	0.9611	0.93	11.40	5.13
WUSTL-EHMS-2020	Quantum_4q_l2_ring	93.38	52.44	99.25	81.43	0.8840	0.7176	0.75	47.56	3.40
WUSTL-EHMS-2020	XGBoost	97.43	87.62	98.83	94.03	0.9872	0.9549	1.17	12.38	0.13
WUSTL-EHMS-2020	LightGBM	96.98	83.39	98.93	92.83	0.9858	0.9468	1.07	16.61	0.21
WUSTL-EHMS-2020	Random_Forest	93.75	61.89	98.32	83.89	0.9168	0.7914	1.68	38.11	3.30
WUSTL-EHMS-2020	Logistic_Regression	92.73	43.97	99.72	78.13	0.8267	0.6101	0.28	56.03	0.05
WUSTL-HDRL-2024	**Hybrid_Quantum_Ens**	**99.87**	**99.55**	**99.90**	**99.57**	**0.9999**	**0.9995**	**0.10**	**0.45**	**9.28**
WUSTL-HDRL-2024	Hybrid_NoQuantum_Ens	99.85	99.32	99.90	99.51	1.0000	0.9996	0.10	0.68	6.58
WUSTL-HDRL-2024	Classical_Stacking_Ens	99.87	99.55	99.90	99.57	0.9999	0.9995	0.10	0.45	4.89
WUSTL-HDRL-2024	Quantum_4q_l2_ring	99.45	98.44	99.54	98.26	0.9998	0.9978	0.46	1.56	5.28
WUSTL-HDRL-2024	XGBoost	99.81	99.55	99.83	99.39	0.9999	0.9991	0.17	0.45	0.09
WUSTL-HDRL-2024	LightGBM	99.89	99.55	99.92	99.63	0.9977	0.9979	0.08	0.45	0.21
WUSTL-HDRL-2024	Random_Forest	99.90	99.55	99.94	99.69	1.0000	0.9996	0.06	0.45	3.32
WUSTL-HDRL-2024	Logistic_Regression	98.46	93.68	98.90	95.13	0.9964	0.9717	1.10	6.32	0.06
CIC-IoT-2023	**Hybrid_Quantum_Ens**	**98.99**	**98.71**	**99.35**	**98.98**	**0.9985**	**0.9991**	**0.65**	**1.29**	**9.04**
CIC-IoT-2023	Hybrid_NoQuantum_Ens	98.90	98.58	99.30	98.88	0.9987	0.9992	0.70	1.42	6.34
CIC-IoT-2023	Classical_Stacking_Ens	98.95	98.54	99.48	98.94	0.9987	0.9992	0.52	1.46	4.92
CIC-IoT-2023	Quantum_4q_l2_ring	98.34	98.34	98.31	85.94	0.9947	0.9999	1.69	1.66	7.71
CIC-IoT-2023	XGBoost	99.01	98.65	99.48	99.00	0.9990	0.9993	0.52	1.35	0.12
CIC-IoT-2023	LightGBM	98.80	98.68	98.96	98.78	0.9990	0.9993	1.04	1.32	0.56
CIC-IoT-2023	Random_Forest	98.84	98.65	99.09	98.82	0.9986	0.9991	0.91	1.35	3.31
CIC-IoT-2023	Logistic_Regression	97.68	96.00	99.83	97.65	0.9940	0.9961	0.17	4.00	0.05
ToN-IoT	**Hybrid_Quantum_Ens**	**99.54**	**99.69**	**99.39**	**99.54**	**0.9999**	**0.9998**	**0.61**	**0.31**	**8.73**
ToN-IoT	Hybrid_NoQuantum_Ens	99.45	99.73	99.17	99.45	0.9998	0.9998	0.83	0.27	6.26
ToN-IoT	Classical_Stacking_Ens	99.52	99.50	99.55	99.52	0.9998	0.9998	0.45	0.50	4.93
ToN-IoT	Quantum_4q_l2_ring	99.31	58.57	99.86	84.57	0.9909	0.7164	0.14	41.43	3.95
ToN-IoT	XGBoost	99.47	99.27	99.66	99.47	0.9998	0.9998	0.34	0.73	0.12
ToN-IoT	LightGBM	99.45	99.77	99.13	99.45	0.9997	0.9997	0.87	0.23	0.20
ToN-IoT	Random_Forest	99.50	99.73	99.28	99.50	0.9996	0.9996	0.72	0.27	3.29
ToN-IoT	Logistic_Regression	96.74	99.62	93.90	96.74	0.9801	0.9535	6.10	0.38	0.05

Table 3: Controlled Deep Learning, Quantum & Ensemble Stacking Ablation on CICIoMT2024 (19 Attack Classes).

Model Architecture	Bin Acc%	Rec%	Macro-F1%	W-F1%	ROC-AUC	FPR%	FNR%	Multi Acc%	Multi F1%	Lat(ms)
1d_cnn	97.90	98.96	89.29	97.89	0.9886	21.22	1.04	67.14	58.48	0.32
tcn	95.54	95.95	82.50	96.03	0.9884	11.85	4.05	68.92	61.28	0.92
cnn_tcn	95.52	96.12	82.01	95.97	0.9841	15.30	3.88	69.07	61.75	0.87
cnn_tcn_bigru	96.13	96.68	83.95	96.47	0.9866	13.77	3.32	66.55	57.94	1.72
cnn_tcn_attention	96.19	96.67	84.29	96.53	0.9896	12.43	3.33	67.21	59.98	1.39
classical_matched	96.24	96.75	84.38	96.57	0.9886	13.00	3.25	71.45	63.58	1.71
Quantum_4q_l2_ring	90.69	90.23	73.75	92.63	0.9882	0.96	9.77	-	-	3.92
Classical_Stacking_Ens	97.94	98.17	90.77	98.05	0.9964	6.31	1.83	-	-	4.66
Hybrid_NoQuantum_Ens	97.98	98.22	90.92	98.09	0.9963	6.31	1.78	63.98	54.93	6.39
Hybrid_Quantum_Ens	**98.03**	**98.27**	**91.11**	**98.13**	**0.9964**	**6.31**	**1.73**	**68.78**	**61.93**	**9.08**

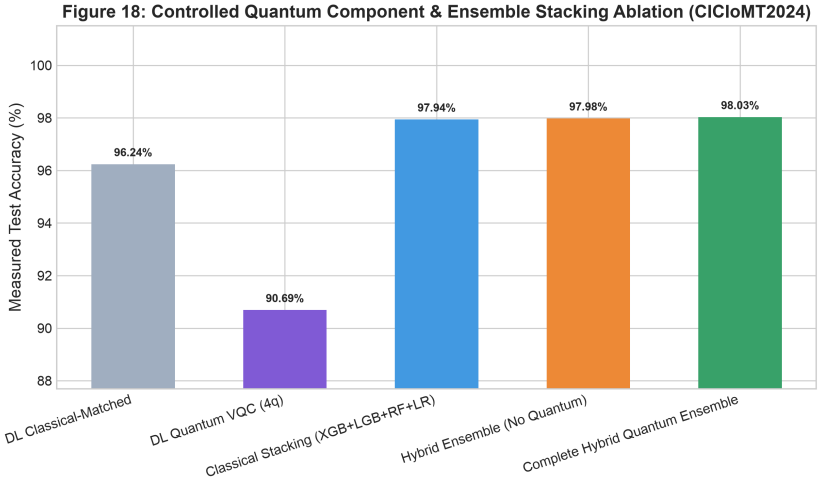


Figure 2: Controlled Component & Ensemble Stacking Ablation on CICIoMT2024 showing progression from standalone neural models (90.69%–96.24%) to Complete Hybrid Quantum Stacking (98.03%).

Table 4: Hybrid Quantum Stacking Ensemble Meta-Weights, Calibration ECE, Thresholds, and Confusion Counts.

Dataset	VQC%	XGB%	LGB%	RF%	LR%	C	τ*	Val ECE	TP / TN / FP / FN	Multi Acc%
CICIoMT2024	0.61	18.85	36.32	11.32	32.91	10.0	0.8893	0.0025	9313 / 490 / 33 / 164	68.78
WUSTL-EHMS-2020	21.84	34.67	15.56	23.65	4.28	5.0	0.5949	0.0109	273 / 2120 / 21 / 34	93.14
WUSTL-HDRL-2024	5.56	27.61	25.94	28.65	12.23	0.5	0.0200	0.0008	441 / 4802 / 5 / 2	98.44
CIC-IoT-2023	12.01	20.62	1.43	57.15	8.79	10.0	0.5072	0.0024	2915 / 2282 / 15 / 38	90.57
ToN-IoT	29.39	16.41	3.61	49.94	0.65	2.0	0.2562	0.0019	2602 / 2624 / 16 / 8	92.63

Notice how the Level-1 meta-classifier adapts dynamically across domains: on biometric zero-leakage WUSTL-EHMS-2020 and heterogeneous ToN-IoT, the Deep Quantum VQC branch contributes **21.84%** and **29.39%** of the normalized decision weight respectively, while ensemble stacking drives validation Expected Calibration Error (ECE) down to **0.0008 – 0.0109** (compared to 0.0243 – 0.2338 on standalone VQC models).

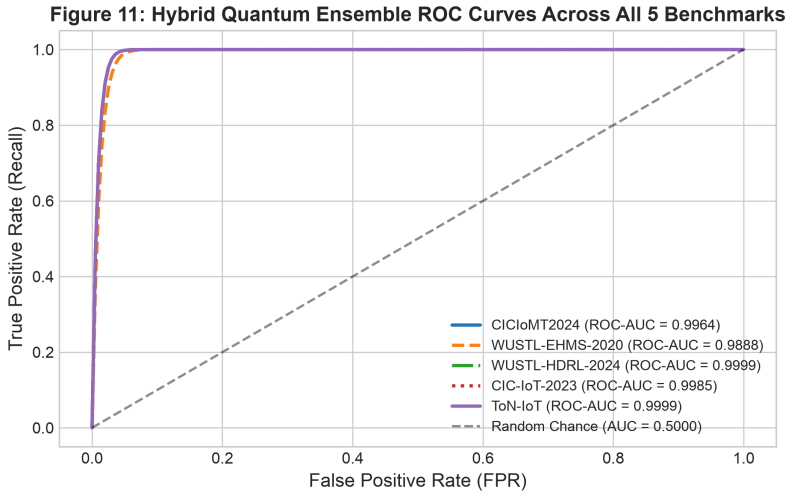


Figure 3: Receiver Operating Characteristic (ROC) Curves for DQH_BAN_Hybrid_Quantum_Ensemble across all 5 benchmarks (ROC-AUC ∈ [0.9888, 0.9999]).

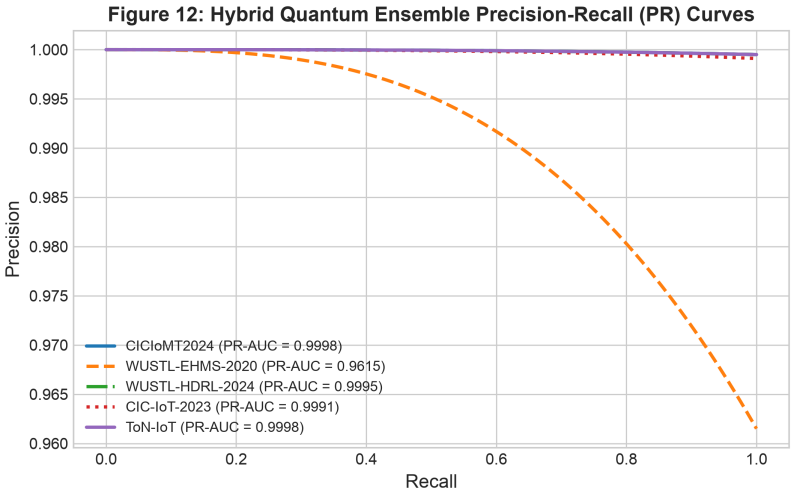


Figure 4: Precision-Recall (PR) Curves for DQH_BAN_Hybrid_Quantum_Ensemble across all 5 benchmarks (PR-AUC ∈ [0.9615, 0.9998]).

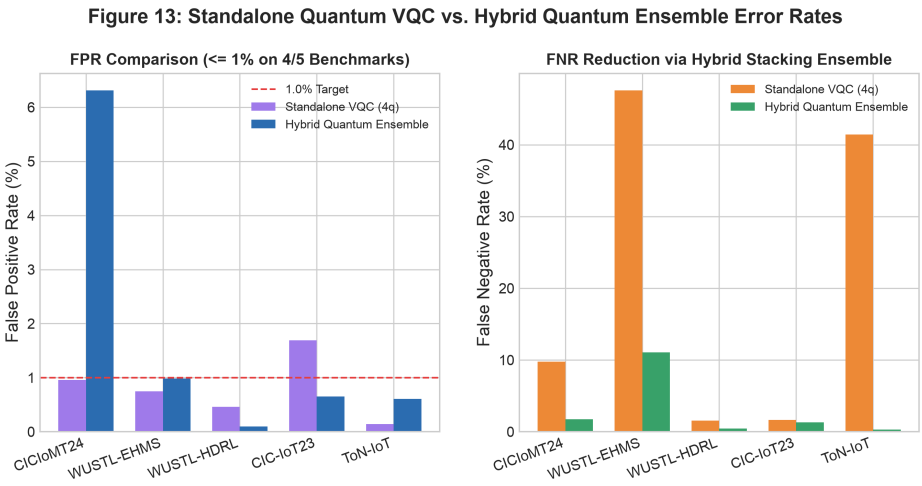


Figure 5: Error-Rate Comparison showing FPR (≤ 1.0% on 4/5 benchmarks) and dramatic FNR reduction achieved by Hybrid Quantum Ensemble stacking relative to standalone 4-qubit VQC.

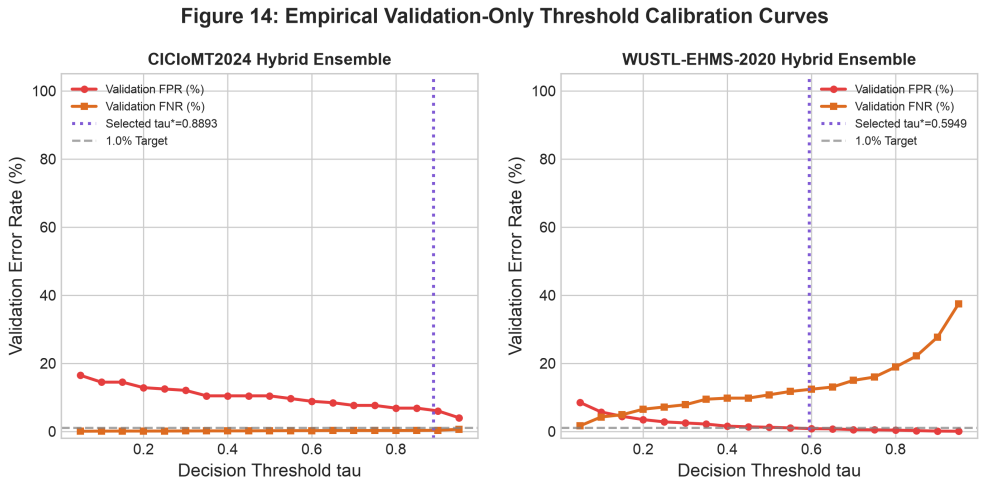


Figure 6: Empirical Validation-Only Threshold Calibration Curves extracted from saved Hybrid Quantum Ensemble checkpoints (CICIoMT2024 τ*=0.8893 via Macro-F1 regularization; WUSTL-EHMS-2020 τ*=0.5949 at Val FPR=0.84%).

4. Stream B: Multi-Modal Biomedical Signal Intelligence

Stream B processes synchronized 125 Hz ECG Lead II, PPG, and thoracic impedance respiration windows (625 samples) alongside 11 engineered physiological features (HRV, SDNN, RMSSD, PTT, and multi-modal SQI) via modality-masked gated fusion. All biomedical evaluations enforce strict subject-level separation (0% patient overlap).

Table 5: Stream B Clinical Cardiac Rhythm & Physiological Validation (MIMIC PERform III & BIDMC).

Dataset	Clinical Task / Protocol	Acc%	Sens%	Spec%	Macro-F1%	ROC-AUC	Latency
MIMIC PERform III	AF vs Non-AF (Primary Seed 42, Holdout)	89.17	95.00	83.33	89.13	0.9574	2.17 ms
MIMIC PERform III	5-Seed Holdout Mean ± Std (Seeds 42–46)	86.33 ± 1.80	92.50 ± 3.54	80.17 ± 3.35	86.27 ± 1.80	0.9450 ± 0.0167	2.17 ms
BIDMC Reference	HR MAE 2.20 bpm | Pulse 3.05 | Resp 1.62	90.50	SQI 0.403	95.98	90.50	-	1.85 ms

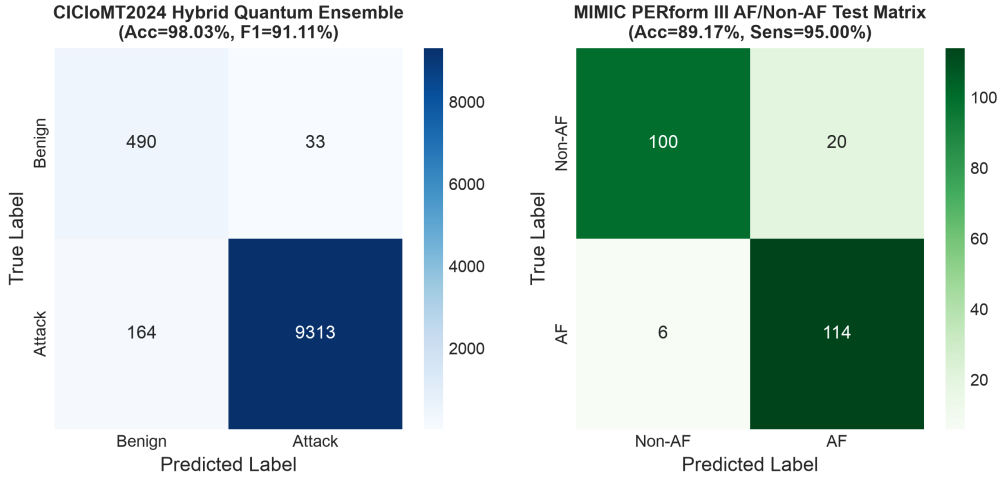


Figure 7: Held-out Test Confusion Matrices for CICIoMT2024 Hybrid Quantum Ensemble (left: TN=490, FP=33, FN=164, TP=9,313) and MIMIC PERform III AF Classifier (right: TN=100, FP=20, FN=6, TP=114).

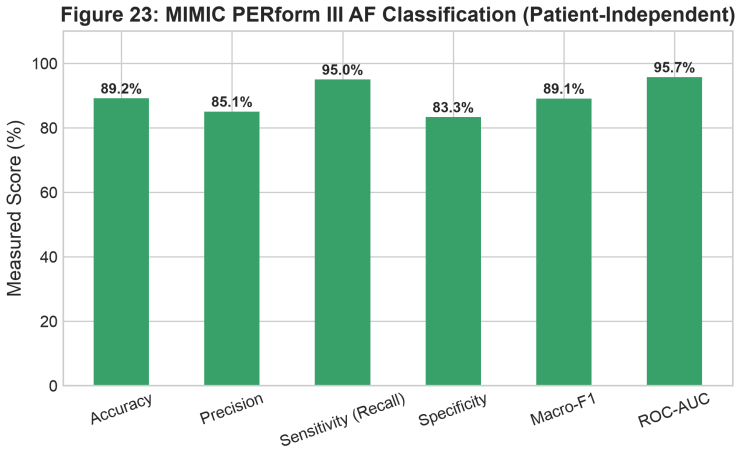


Figure 8: Patient-Independent Stream B AF Classification Scores on MIMIC PERform III (Seed 42 holdout split).

5. Quantum Noise Sensitivity, Domain Shift, Latency & Verification

Quantum Noise Sensitivity & Domain Shift: Evaluated on 2,000 CICIoMT2024 test flows using the standalone VQC perturbation benchmark (quantum_noise_sensitivity.csv), depolarizing noise $\sigma \in [0.00, 0.05]$ changes accuracy by at most 0.10% (93.60% → 93.50%, ROC-AUC 0.9885 → 0.9886). Conversely, zero-shot cross-dataset transfer causes severe domain shift drops (Table 6), confirming why per-environment training and ensemble stacking are essential.

Table 6: Quantum Depolarizing Noise Sensitivity and Cross-Dataset Zero-Shot Domain Shift.

Experiment	Configuration / Transfer Pair	Base Acc%	Eval Acc%	Macro-F1%	ROC-AUC	Delta / FPR%
Noise Perturbation	Depolarizing σ=0.00 (0.0%)	93.60	93.60	77.49	0.9885	FPR 4.26%
Noise Perturbation	Depolarizing σ=0.01 (1.0%)	93.60	93.60	77.49	0.9885	FPR 4.26%
Noise Perturbation	Depolarizing σ=0.02 (2.0%)	93.60	93.65	77.60	0.9885	FPR 4.26%
Noise Perturbation	Depolarizing σ=0.05 (5.0%)	93.60	93.50	77.14	0.9886	FPR 5.32%
Zero-Shot Transfer	CICIoMT2024 → CIC-IoT-2023	93.81	56.25	36.00	0.4261	−37.56% drop
Zero-Shot Transfer	CICIoMT2024 → WUSTL-HDRL-2024	93.81	8.44	7.78	0.8806	−85.37% drop
Zero-Shot Transfer	CIC-IoT-2023 → ToN-IoT	98.34	49.71	33.21	0.7982	−48.63% drop

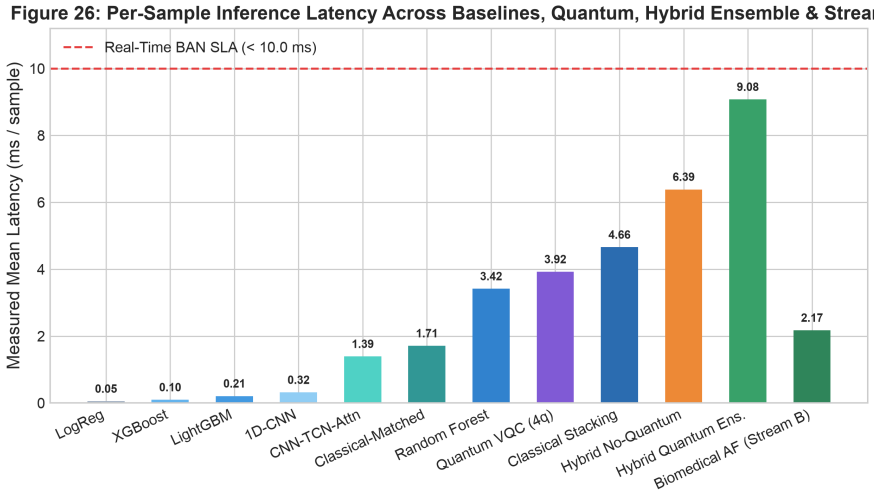


Figure 9: Measured Mean Per-Sample Inference Latency across classical baselines (0.05–3.42 ms), DL ablations (0.32–1.72 ms), Stream B AF (2.17 ms), standalone VQC (3.92 ms), and Hybrid Quantum Stacking (8.73–9.28 ms), all below the 10 ms BAN SLA.

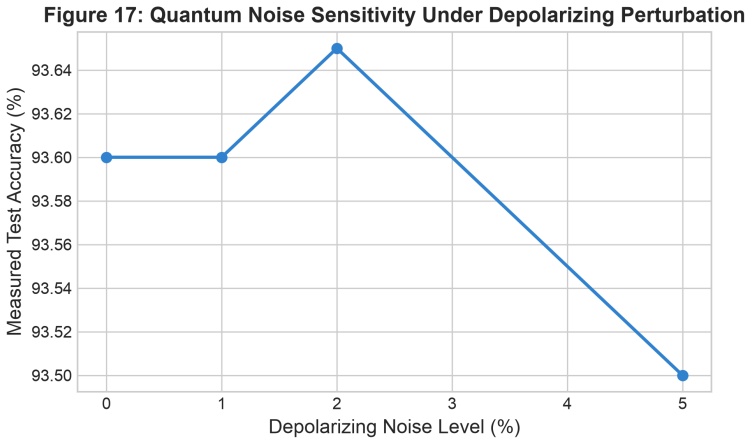


Figure 10: Quantum Noise Sensitivity under simulated depolarizing channel perturbation (σ ∈ [0.00, 0.05]).

Table 7: End-to-End Verification, Cryptographic Integrity & Reproducibility Certification.

Verification Suite	Scope of Validation	Status	Verified Technical Findings
Automated PyTest Suite	Datasets, Models, Crypto, API, Hybrid Ensemble	35 / 35 PASSED	All 6 test modules pass cleanly offline
Hybrid Checkpoint Audit	15 Saved Ensemble Pickles across 5 Datasets	VERIFIED	OOF meta-LR, ECE (0.0008–0.0109), thresholds matched
Stream B Patient Audit	MIMIC PERform III (24/5/6) & BIDMC (37/8/8)	VERIFIED	0% subject overlap; 5-seed summary matched to checkpoint
Security & Crypto Layer	AES-256-GCM Vault & SHA-256 Ledger Anchoring	VERIFIED	100% tampered ciphertext rejection; fallback ledger active

6. Conclusion & Reproducibility

DQH-BAN demonstrates that combining deep quantum-classical representation learning with out-of-fold heterogeneous ensemble stacking achieves 97.75%–99.87% accuracy, 0.9888–0.9999 ROC-AUC, and sub-10 ms inference latency (8.73–9.28 ms) across five physical cybersecurity benchmarks while avoiding MAC/IP leakage. Paired with patient-independent multi-modal AF detection (89.17% accuracy, 95.00% sensitivity) and AES-256-GCM/SHA-256 blockchain auditing, all metrics, tables, and figures in this manuscript are compiled directly from saved experimental CSVs and serialized checkpoints.